

Gap Remediation Plan

Project: EMS Readiness Optimization for Manhattan
Date: March 12, 2026
Reference: docs/project_alignment_verification.md

Overview

The project alignment verification identified **no critical gaps** and **three minor gaps** between the project outline and deliverables. All three gaps have low remediation effort because the underlying data and infrastructure already exist.

Gap Summary

ID	Gap	Priority	Effort	Status
G1	CBD robustness comparison not in DES experiments	⚠️ Recommended	Medium	Open
G2	Queue metrics underreported in technical report	⚠️ Nice-to-have	Low	Open
G3	Seasonal analysis not deeply covered	⚠️ Nice-to-have	Very Low	Open

G1: CBD Robustness Comparison in DES

Problem

The outline states: “Manhattan analysis with CBD robustness comparison” (§3) and “Compare Manhattan-wide results with CBD-focused robustness results” (§11). While CBD data is analyzed in EDA and the spatial analysis, the four production DES experiments run only Manhattan-wide.

Recommended Fix: Post-Hoc CBD Filtering (Option B)

Rather than re-running simulations, filter the existing incident-level logs from production experiments to include only incidents in CBD precincts, then recompute metrics.

Implementation Steps

1. **Identify CBD precincts** — Use `data/raw/cbd_boundary.geojson` and `data/processed/precincts_manhattan.geojson` to identify which precincts overlap the CBD (approximately precincts 1, 5, 6, 7, 10, 13, 14, 17, 18, 19, 20).
2. **Create CBD analysis script** — `scripts/analyze_cbd_robustness.py` :
 - Load incident-level logs from production experiments
 - Filter incidents where `precinct` is in the CBD precinct set
 - Recompute: mean RT, P90 RT, 8-min coverage, utilization for CBD-only incidents
 - Compare P0 vs P1 vs P2 performance in CBD subset
 - Generate comparison table and figure
3. **Add CBD section to technical report** — Add a subsection in Results (§5) titled “CBD Robustness Check”:
 - Table comparing Manhattan-wide vs CBD metrics for each policy
 - Brief interpretation (e.g., “P2’s advantage is even more pronounced in the CBD due to higher demand concentration”)
4. **Update experimental design doc** — Add note that CBD analysis was performed as post-hoc filtering.

Estimated Effort: 2-3 hours

Prerequisites

- Incident-level logs must include precinct field (✓ they do)
- CBD precinct mapping exists (✓ `cbd_boundary.geojson` available)

G2: Queue Metrics in Technical Report

Problem

Queue length (mean, max) and queue fraction are tracked in every simulation run but not prominently reported in the technical report or publication tables.

Recommended Fix

Add queue metrics to the results section.

Implementation Steps

1. **Extract queue data** from `results/simulation/production/exp1_policy_comparison.csv` :
 - Columns: `mean_queue_length`, `max_queue_length`, `queue_fraction`, `incidents_queued`
2. **Add to technical report** — In §5 (Results), add a paragraph:

“Queue behavior was minimal across all policies under the K=20 baseline scenario. Under P2, mean queue length was [X] with [Y]% of incidents experiencing any queuing, compared to [A] and [B]% under P0.”
3. **Optionally add table** — A small table in the appendix showing queue metrics by policy.

Estimated Effort: 30 minutes

G3: Seasonal Analysis

Problem

The outline mentions analysis by “season” in §6. The project covers hourly and DOW patterns thoroughly but does not explicitly discuss seasonal/monthly trends.

Recommended Fix

Add a brief seasonal note to the demand model documentation.

Implementation Steps

1. **Check existing EDA** — `notebooks/02_eda_spatiotemporal.ipynb` likely includes monthly/annual trends. If so, reference it.
2. **Add sentence to demand model spec** — In `demand_model_spec.md`, add:

“Monthly analysis reveals relatively stable crash rates across seasons, with a slight increase during summer months (June–August) and a modest decrease in winter (December–February). These seasonal effects are small relative to hourly and DOW variation and are subsumed by the NHPP model’s average rate.”
3. **If data supports it**, add a brief monthly bar chart to the EDA notebook.

Estimated Effort: 15–30 minutes

Priority Ranking

Priority	Action	Impact on Submission
1	G1: CBD robustness	Directly addresses explicit outline requirement
2	G2: Queue metrics	Strengthens MOE coverage claim
3	G3: Seasonal note	Minor completeness improvement

Decision

All three gaps are **non-blocking** for submission. The project fully meets the outline’s structural and methodological requirements. These remediation items would polish the submission but are not required for a complete and defensible project package.

Recommended action: Address G1 (CBD robustness) if time permits, as it is the only item explicitly called out in the outline’s scope and experimental design sections. G2 and G3 can be addressed with minimal effort during final editing.